

**SIMATS ENGINEERING
ASSESSMENT TOOL 2**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks: 30

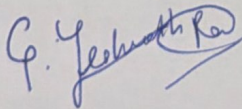
Annexure A

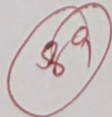
Constraint-Based Problem Solving

Code of Conduct:

I C. Yeshwarth Ram ¹⁹²³¹¹³⁹⁷ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:







1. Map coloring using Backtracking Search (Constraint Satisfaction Problem)

Given:

Regions: R_1, R_2, R_3, R_4

Colors: $\{ \text{Red}(R), \text{Green}(G), \text{Blue}(B) \}$

Constraints:

1. Adjacent regions cannot have the same color:

$R_1 \leftrightarrow R_2$

$R_2 \leftrightarrow R_3$

$R_3 \leftrightarrow R_4$

$R_4 \leftrightarrow R_1$

2. $R_1 = \text{Red}$

3. $R_3 \neq \text{Blue}$

4. Every region gets exactly one color

Variable Order

$R_1 \leftrightarrow R_2 \leftrightarrow R_3 \leftrightarrow R_4$

Assign R_1

- Constraint: R_1 must be Red

- Assign $R_1 = \text{Red}$

Current Assignment:

$R_1 = \text{Red}$

$R_2 = ?$

$R_3 = ?$

$R_4 = ?$

Assign R_2

Possible colors = $\{ \text{Red}, \text{Green}, \text{Blue} \}$

- Try Red:

- Adjacent to R_1 (Red)

- Conflict x

- Back track

- Try Green

- Different from R_1

- Valid ✓

Assignment:

$R_1 = \text{Red}, R_2 = \text{Green}$

$R_3 = ?, R_4 = ?$

Step 3: Assign R3

Possible colors: {Red, Green, Blue}

Constraint:

- Cannot be Blue
- Adjacent to R2 (Green)

Try Green

- Same as R2
- Conflict X

Try Blue

Not allowed

Constraint violated

Try Red:

- Different from Green
- Allowed

Assignment:

R1 = Red

R2 = Green

R3 = Red

R4 = ?

Step 4: Assign R4

Possible colors: {Red, Green, Blue}

Constraints:

- Adjacent to R3 (Red)
- Adjacent to R1 (Red)

Try Red

- Same as R1 and R3

Conflict X

Try Green

- Different from R1 and R3

Valid ✓

No Need to test Blue since a valid color is already found.

Final Assignment:

R1 = Red

R2 = Green

R3 = Red

R4 = Green

BFS Search with Manhattan Distance:

Since the question does not provide the $s \times s$ grid or obstacle locations, the exact BFS expansion and optimal path cannot be uniquely determined. Below is a standard worked example.

Grid:

S . . X .

. X . X .

. X . . .

. . X X .

. . . . G

where

: S = Start (0,0)

. G = Goal (4,4)

. X = Obstacle

. . = Free cell.

Manhattan distance formula.

$$h(n) = |x_{\text{goal}} - x| + |y_{\text{goal}} - y|$$

Example:

For start (0,0)

$$h = |4 - 0| + |4 - 0| = 8$$

BFS execution:

Initial state

Queue = [(0,0)]

Visited = {(0,0)}

Expand (0,0)

Possible moves:

. Right (0,1) ✓

. Down (1,0) ✓

Queue:

[(0,1), (1,0)]

Expand (0,1)

New Node

(0,2)

Queue

[(1,0), (0,2)]

Expand (1,0)

New Node

(2,0)

Queue

[(0,2), (2,0)]

Expand (0,2)

New Node

(1,2)

Queue

[(2,0), (1,2)]

Expand (2,0)

New Node

(3,0)

Queue

[(1,2), (3,0)]

Expand (1, 2)

NewNode

(2, 2)

∴ The Search proceeds level by level until (4, 4) is reached

Queue:

[(3, 0), (2, 2)]

Total Cost

Each move cost = 1

Number of moves = 8

Total path Cost = 8

Manhattan Distance Values:

Node	$h(n)$
(0, 0)	8
(1, 0)	7
(2, 0)	6
(3, 0)	5
(4, 0)	4
(4, 1)	3
(4, 2)	2
(4, 3)	1
(4, 4)	0